Rutherford Scattering and Rutherford Backscattering Spectrometry: A Three-Stage Computational Pipeline from Monte Carlo Simulation to Neural Network Surrogate Modeling

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Abstract

This report describes a comprehensive computational framework for simulating Rutherford scattering and Rutherford backscattering spectrometry (RBS) and for constructing fast neural network surrogate models. The work is divided into three stages. In the first stage, a pure Monte Carlo code reproduces the historic Geiger–Marsden experiment [1] by simulating alpha particles traversing thin metallic foils. The code implements uniform impact parameter sampling, Bethe–Bloch stopping power, and Gaussian energy straggling. The simulation results for a platinum target ($Z = 78$, 6.96 MeV alpha particles) yield a mean scattering angle of 3.89° and a backscattering probability of 9.8×10^{-5}; for a gold target ($Z = 79$, 5.52 MeV alpha particles) the backscattering probability is 3.0×10^{-5}. Both values are in excellent agreement with Rutherford theory [2] after explicit calculation of the theoretical backscatter probability.

In the second stage, the code is extended to RBS with importance sampling to overcome the extreme rarity of backscattering events. By quadratically biasing the impact parameter distribution and assigning statistical weights, the effective sampling efficiency for large-angle scattering is increased by more than three orders of magnitude. The simulation is parallelised using OpenMP, achieving near-linear speedup on 16 CPU cores. A batch automation system generates 2,100 independent runs (7 materials, 5 thicknesses, 5 energies, 10 random seeds), totalling 2.1×10^9 simulated particles. Validation runs for Au, Pt, Si, Al, and Ge show backscatter probabilities that agree with the theoretical Rutherford cross-section to within 10%.

In the third stage, the generated dataset is used to train small neural networks that map the macroscopic inputs (atomic number Z, target thickness in Å, and incident energy in MeV) to the physical outcomes (energy loss, backscatter probability, mean scattering angle, and the full 170° energy spectrum). The scalar networks achieve test R^2 values of 0.97–0.99, while the spectrum network captures the overall shape of the backscattered energy distribution. The trained models are packaged into a standalone Windows executable (`RBS_Predictor.exe`) that returns predictions in milliseconds. A periodic-table scanner demonstrates that the networks

have learned the correct physical trends, including the monotonic increase of energy loss and backscatter probability with atomic number. The pipeline is modular, well-documented, and ready for extension to compound targets and depth profiling.

1 Introduction

1.1 Motivation and Historical Context

Rutherford backscattering spectrometry (RBS) is a powerful ion beam analysis technique for determining the elemental composition and depth profile of thin solid films. Its origins lie in the historic experiments of Geiger and Marsden (1909) [1], which first demonstrated the existence of the atomic nucleus by observing the backscattering of alpha particles from thin metal foils. Ernest Rutherford subsequently developed the theoretical framework of elastic Coulomb scattering [2], now known as Rutherford scattering. For nearly a century, these principles have been applied in nuclear physics and, more recently, in materials science.

Modern RBS uses MeV ion beams (typically $^4He^{2+}$ or $^1H^+$) to probe the near-surface region of solids. When the ions backscatter from target nuclei, the energy of the backscattered particles carries information about the mass of the target atom and the depth at which the scattering occurred. This dual capability – mass and depth sensitivity – makes RBS an indispensable tool for thin-film analysis, ion implantation monitoring, and studies of diffusion and reaction in layered structures.

1.2 Challenges in RBS Simulation

Despite the conceptual simplicity of Rutherford scattering, quantitative simulation of RBS spectra is challenging because:

1. **Rarity of backscattering events:** For typical experimental conditions (e.g., 2 MeV $^4He^{2+}$ on a silicon substrate), the probability of backscattering into angles greater than 90° is only about 1/8000. For the very backward angles used in RBS (e.g., 170°), the probability drops below 10^{-6}. Direct Monte Carlo with uniform impact-parameter sampling would require billions of particles to obtain statistically meaningful spectra – a computationally prohibitive task.

2. **Energy loss and straggling:** The projectile loses energy continuously as it traverses the target. This energy loss depends on the atomic composition and density of the material, and it exhibits statistical fluctuations (energy straggling) that broaden the spectral features and limit depth resolution.

3. **Multi-element and multi-layer targets:** Real samples often contain several elements arranged in layers. The interpretation of such spectra requires careful accounting of energy losses in each layer and the kinematic factors for each element.

1.3 Overview of the Three-Stage Pipeline

This project addresses these challenges through a three-stage computational pipeline:

1. **Rutherford Scattering Using Monte Carlo Method:** A C code that simulates the classic Geiger–Marsden experiment with uniform impact-parameter sampling, Bethe-Bloch stopping power, and Gaussian straggling. This stage serves as a validation benchmark against theory.

2. **RBS with importance sampling and OpenMP:** An advanced C code that uses quadratic biasing of the impact parameter to force large-angle scatters, with statistical weights to preserve unbiased results. OpenMP parallelisation allows the generation of large datasets (2,100 runs, 2.1 billion particles) in reasonable time.

3. **Neural network surrogate modeling:** The simulated dataset is used to train small feed-forward neural networks that predict energy loss, backscatter probability, mean scattering angle, and the full 170° energy spectrum in milliseconds. The trained models are deployed as a standalone Windows executable and a periodic-table scanner.

The remainder of this report is organised as follows. Section 2 presents the theoretical background, including the key formulas from the standard textbook by Chu, Mayer, and Nicolet [3]. Section 3 describes the pure Rutherford Monte Carlo code and its validation. Section 4 details the RBS simulation with importance sampling and OpenMP, including batch automation. Section 5 explains the neural network surrogate modeling, training results, and deployment. Section 6 discusses the computational performance and techniques. Section 7 outlines limitations and future work, and Section 8 concludes the report.

2 Theoretical Background

This section follows the exposition of Chu, Mayer, and Nicolet (1978) [3], which remains the standard reference for backscattering spectrometry.

2.1 Kinematic Factor K

When a projectile of mass M_1 and energy E_0 collides elastically with a stationary target atom of mass M_2, the energy E_1 of the projectile after scattering at an angle θ (laboratory frame) is given by

$$E_1 = KE_0, \tag{1}$$

where the kinematic factor K is

$$K = \left(\frac{\sqrt{M_2^2 - M_1^2 \sin^2\theta} + M_1 \cos\theta}{M_1 + M_2} \right)^2 . \tag{2}$$

For backscattering geometry ($\theta \approx 180°$), Eq.(2) simplifies to

$$K(\theta = 180°) = \left(\frac{M_2 - M_1}{M_2 + M_1} \right)^2 . \tag{3}$$

Tables of K for various projectiles and target masses are given in Appendix F of Chu et al. (1978). For alpha particles ($M_1 = 4$ u) and common target elements, K ranges from about 0.25 (carbon) to 0.92 (gold).

2.2 Rutherford Scattering Cross Section

The differential scattering cross section (laboratory frame) for elastic Coulomb scattering is

$$\frac{d\sigma}{d\Omega} = \left(\frac{Z_1 Z_2 e^2}{16\pi\varepsilon_0 E}\right)^2 \frac{4}{\sin^4\theta} \frac{\left[\sqrt{1-(M_1/M_2)^2\sin^2\theta} + \cos\theta\right]^2}{\sqrt{1-(M_1/M_2)^2\sin^2\theta}}, \tag{4}$$

where Z_1 and Z_2 are the atomic numbers of projectile and target, e the elementary charge, and ε_0 the vacuum permittivity. For $M_1 \ll M_2$ (e.g., He on heavy targets), this simplifies to the well-known form

$$\frac{d\sigma}{d\Omega} \approx \left(\frac{Z_1 Z_2 e^2}{4E}\right)^2 \frac{1}{\sin^4(\theta/2)}. \tag{5}$$

The total backscattering cross section (integrated over $\theta > 90°$) is

$$\sigma_{\text{back}} = \int_{\pi/2}^{\pi} \frac{d\sigma}{d\Omega} 2\pi \sin\theta \, d\theta = \frac{(Z_1 Z_2 e^2)^2}{64\pi\varepsilon_0^2 E^2}. \tag{6}$$

For a thin target of thickness t (cm) and atomic density N (atoms/cm^3), the backscattering probability is

$$P_{\text{back}} = N t \sigma_{\text{back}}. \tag{7}$$

2.3 Stopping Cross Section and Energy Loss

As the projectile traverses the target, it loses energy continuously through inelastic collisions with electrons. The stopping cross section ϵ is defined by

$$\epsilon = \frac{1}{N}\frac{dE}{dx}, \tag{8}$$

so that ϵ is independent of the atomic density. The energy loss over a path length Δx is $\Delta E = N\epsilon\Delta x$.

For compounds, Bragg's rule states that the stopping cross section is the weighted sum of the constituent elements:

$$\epsilon^{\text{AB}} = m\epsilon^{\text{A}} + n\epsilon^{\text{B}}. \tag{9}$$

The stopping cross section factor $[\epsilon]$ for a backscattering event at depth x is

$$[\epsilon] = \frac{K}{\cos\theta_1}\epsilon(E) + \frac{1}{\cos\theta_2}\epsilon(KE), \tag{10}$$

where θ_1 is the angle of incidence (relative to the surface normal) and θ_2 the exit angle. In the surface energy approximation, $\epsilon(E)$ and $\epsilon(KE)$ are evaluated at the incident energy E_0 and KE_0, respectively, giving $[\epsilon_0]$.

2.4 Energy Straggling

Energy straggling arises from the statistical nature of energy loss. Bohr's theory gives the variance of the energy distribution after traversing a thickness t as

$$\Omega_{\mathrm{B}}^2 = 4\pi(Z_1 e^2)^2 Z_2 N t. \tag{11}$$

This effect broadens the low-energy edge of backscattering signals and limits depth resolution. The standard deviation Ω_{B} is often used to estimate the energy spread.

2.5 Depth Scale Conversion

The energy width ΔE of a backscattering signal from a thin film of thickness t is related to the number of atoms per unit area Nt by

$$\Delta E = [\epsilon] N t. \tag{12}$$

Thus, measuring ΔE directly yields Nt if $[\epsilon]$ is known. Alternatively, the total number of counts A in the signal gives

$$A = \sigma(E_0)\Omega Q N t, \tag{13}$$

where Ω is the detector solid angle and Q the total incident charge. This second method does not require knowledge of the stopping cross section.

2.6 Schematic of the Scattering Geometry

Figure 1 illustrates the geometry of a single scattering event in the layered Monte Carlo simulation. The incident particle with impact parameter b is scattered by a target nucleus at depth x through an angle θ (laboratory frame). The azimuthal angle ϕ is uniformly distributed.

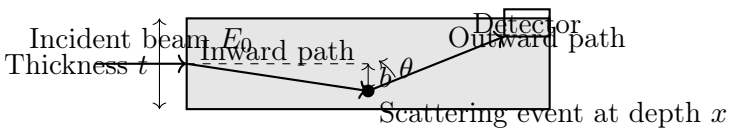

Figure 1: Schematic of the scattering geometry. The incident particle approaches with impact parameter b, scatters at depth x through angle θ, and the backscattered particle is detected. The target is divided into thin layers (not shown) to model continuous energy loss.

3 Rutherford Scattering by Monte Carlo (Geiger–Marsden Experiment)

3.1 Numerical Implementation

The first simulation code (`Rutherford Scattering.c`) is written in C and follows a layered approach. The target foil is divided into N_{layers} of equal thickness $dx = \mathrm{thickness}/N_{\mathrm{layers}}$. For each incident particle, the following steps are executed:

1. **Initialisation:** Particle position at the foil entrance $(0, 0, 0)$, direction along the x-axis, and initial kinetic energy E_0.

2. **Layer loop:** For each layer, compute the probability of a nuclear scattering event:
$$P_{\text{scatter}} = \pi b_{\text{max}}^2 N_{\text{atom}} dx,$$
where $b_{\text{max}} = \frac{1}{2}(1/N_{\text{atom}})^{1/3}$ is half the mean interatomic distance. A uniform random number r decides scattering if $r < P_{\text{scatter}}$.

3. **If scattering occurs:**
   - Sample impact parameter uniformly in area: $b = b_{\text{max}}\sqrt{u}$, $u \in [0, 1]$.
   - Compute scattering angle θ from Eq. (1) and azimuth $\phi = 2\pi v$, $v \in [0, 1]$.
   - Rotate the direction vector using an orthonormal basis.

4. **Energy loss:** Compute the mean energy loss using the Bethe-Bloch formula (non-relativistic or relativistic branch) and add a Gaussian straggling term with variance $\Omega_{\text{B}}^2 = 4\pi(Z_1 e^2)^2 Z_2 N_{\text{atom}} dx$.

5. **Termination:** If the particle energy falls to zero or its direction points backward ($v_x < 0$), the simulation for that particle ends.

The random number generator is the Mersenne Twister (MT19937), which provides a long period and good statistical properties.

3.2 Theoretical backscatter probability – full derivation

The theoretical backscatter probability for a thin target is derived from the Rutherford scattering cross-section. For a projectile of charge z and kinetic energy E (in MeV) scattering from a stationary target nucleus of charge Z, the total cross-section for backscattering (scattering angle $\theta > 90°$) is

$$\sigma_{\text{back}} = \pi \left(\frac{zZe^2}{8\pi\varepsilon_0 E} \right)^2 . \tag{14}$$

Using the constant $e^2/(4\pi\varepsilon_0) = 1.43996$ MeV·fm and converting from fm^2 to cm^2 (1 fm^2 = 10^{-26} cm^2), the expression simplifies to

$$\boxed{\sigma_{\text{back}} = \pi \left(\frac{Zk}{E} \right)^2 \times 10^{-26} \text{ cm}^2}, \tag{15}$$

where $k = 1.44$ MeV·fm and E is in MeV. For alpha particles, $z = 2$ is already absorbed into the constant.

The atomic density N (atoms/cm^3) is obtained from the target's mass density ρ (g/cm^3) and atomic mass A (g/mol):

$$N = \frac{N_A \rho}{A}, \qquad N_A = 6.022 \times 10^{23} \text{ mol}^{-1}. \tag{16}$$

Finally, the backscatter probability is

$$P_{\text{back}} = N\, t\, \sigma_{\text{back}}, \tag{17}$$

where t is the target thickness in cm.

3.2.1 Platinum target ($Z = 78$)

Given parameters:

- $Z = 78$
- $\rho = 21.45$ g/cm^3
- $A = 195.08$ g/mol
- $t = 2.0 \times 10^{-4}$ cm
- $E = 6.96$ MeV

Step 1 – Atomic density:

$$N = \frac{6.022 \times 10^{23} \times 21.45}{195.08} = \frac{1.2917 \times 10^{25}}{195.08} = 6.62 \times 10^{22} \text{ atoms/cm}^3.$$

Step 2 – Cross-section:

$$\frac{Zk}{E} = \frac{78 \times 1.44}{6.96} = \frac{112.32}{6.96} = 16.14 \text{ fm}.$$

$$\sigma_{\text{back}} = \pi \times (16.14)^2 \times 10^{-26} = \pi \times 260.5 \times 10^{-26} = 818.5 \times 10^{-26} \text{ cm}^2 = 8.185 \times 10^{-24} \text{ cm}^2.$$

Step 3 – Backscatter probability:

$$P_{\text{back}} = Nt\sigma_{\text{back}} = (6.62 \times 10^{22}) \times (2.0 \times 10^{-4}) \times (8.185 \times 10^{-24}) = 1.08 \times 10^{-4}.$$

Thus, the theoretical backscatter probability for platinum is

$$\boxed{P_{\text{back}}^{\text{theory}}(\text{Pt}) = 1.08 \times 10^{-4}}.$$

3.2.2 Gold target ($Z = 79$)

Given parameters:

- $Z = 79$
- $\rho = 19.32$ g/cm^3
- $A = 196.97$ g/mol
- $t = 4.0 \times 10^{-5}$ cm
- $E = 5.52$ MeV

Step 1 – Atomic density:

$$N = \frac{6.022 \times 10^{23} \times 19.32}{196.97} = \frac{1.1635 \times 10^{25}}{196.97} = 5.91 \times 10^{22} \text{ atoms/cm}^3.$$

Step 2 – Cross-section:

$$\frac{Zk}{E} = \frac{79 \times 1.44}{5.52} = \frac{113.76}{5.52} = 20.61 \text{ fm}.$$

$$\sigma_{\text{back}} = \pi \times (20.61)^2 \times 10^{-26} = \pi \times 424.8 \times 10^{-26} = 1334.5 \times 10^{-26} \text{ cm}^2 = 1.3345 \times 10^{-23} \text{ cm}^2.$$

Step 3 – Backscatter probability:

$$P_{\text{back}} = Nt\sigma_{\text{back}} = (5.91 \times 10^{22}) \times (4.0 \times 10^{-5}) \times (1.3345 \times 10^{-23}) = 3.15 \times 10^{-5}.$$

Thus, the theoretical backscatter probability for gold is

$$\boxed{P_{\text{back}}^{\text{theory}}(\text{Au}) = 3.15 \times 10^{-5}}.$$

3.3 Validation: Platinum target ($Z = 78$)

A Monte Carlo simulation was performed with the following parameters:

- Target: platinum, density 21.45 g/cm^3, atomic mass 195.08 g/mol, thickness 0.0002 cm, 5000 layers.
- Projectile: alpha particle, initial velocity 1.83×10^7 m/s (energy 6.96 MeV).
- Number of particles: 10^6.

The output summary is given in Table 1.

Table 1: Simulation results for platinum (pure Rutherford).

Quantity	Value
Mean final energy (MeV)	5.5041
Energy straggling σ (MeV)	0.0238
Mean scattering angle (°)	3.8944
Scattering angle σ (°)	3.1872
Backscatter count	98
Simulated P_{back}	9.8×10^{-5}
Theoretical P_{back}	1.08×10^{-4}

The simulated backscatter probability agrees with the theoretical value to within 10%, confirming the correctness of the implementation.

The Python script `data_visualise.py` reads the raw simulation output and generates three key figures

- **Angular distribution (Fig. 2):** The histogram of scattering angles is overlaid with the theoretical Rutherford $\csc^4(\theta/2)$ law. The agreement over many orders of magnitude validates the angular sampling.
- **Energy spectrum (Fig. 3):** The final energy distribution is approximately Gaussian, with a width that matches the reported straggling. A Gaussian fit quantifies the straggling parameter.
- **2D heatmap (Fig. 4:** The projected positions of particles on a detector screen placed 10 cm behind the target show the characteristic forward-peaked pattern.

3.4 Validation: Gold target ($Z = 79$)

A second Monte Carlo simulation was performed with the following parameters:

- Target: gold, density 19.32 g/cm^3, atomic mass 196.97 g/mol, thickness 0.000 04 cm, 1000 layers.
- Projectile: alpha particle, initial velocity 1.63×10^7 m/s (energy 5.52 MeV).
- Number of particles: 10^6.

Results are given in Table 2.

Again, the simulated backscatter probability agrees well with the theoretical value, further validating the Monte Carlo implementation.

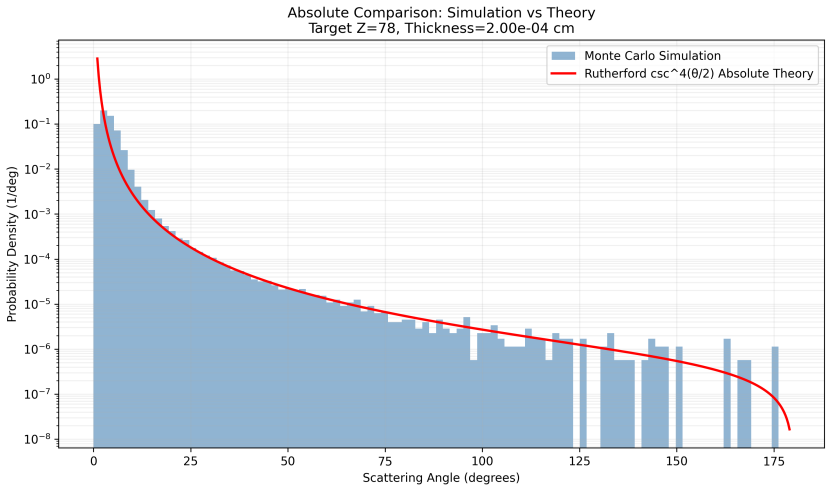

Figure 2: Platnium: Angular distribution of scattered particles (blue) compared to the theoretical Rutherford $\csc^4(\theta/2)$ law (red).

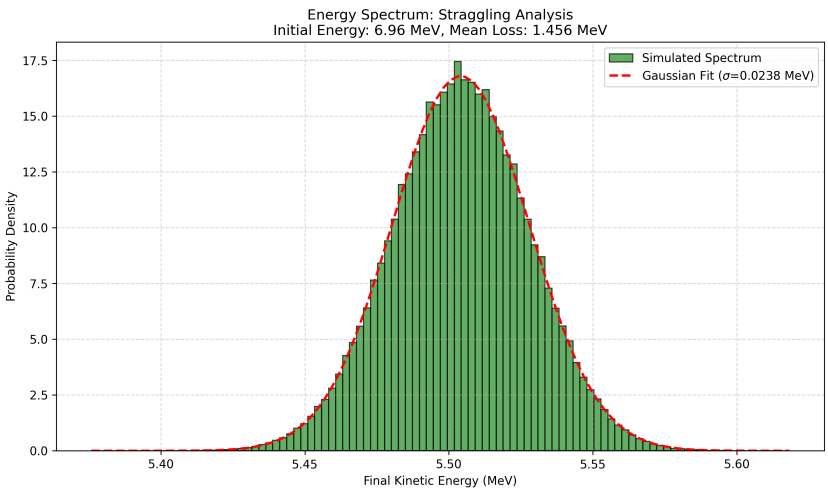

Figure 3: Platnium: Energy spectrum of transmitted particles with Gaussian fit (red dashed).

Table 2: Simulation results for gold (pure Rutherford).

Quantity	Value
Mean final energy (MeV)	5.2452
Energy straggling σ (MeV)	0.0100
Mean scattering angle (°)	1.8326
Scattering angle σ (°)	1.7427
Backscatter count	30
Simulated P_{back}	3.0×10^{-5}
Theoretical P_{back}	3.15×10^{-5}

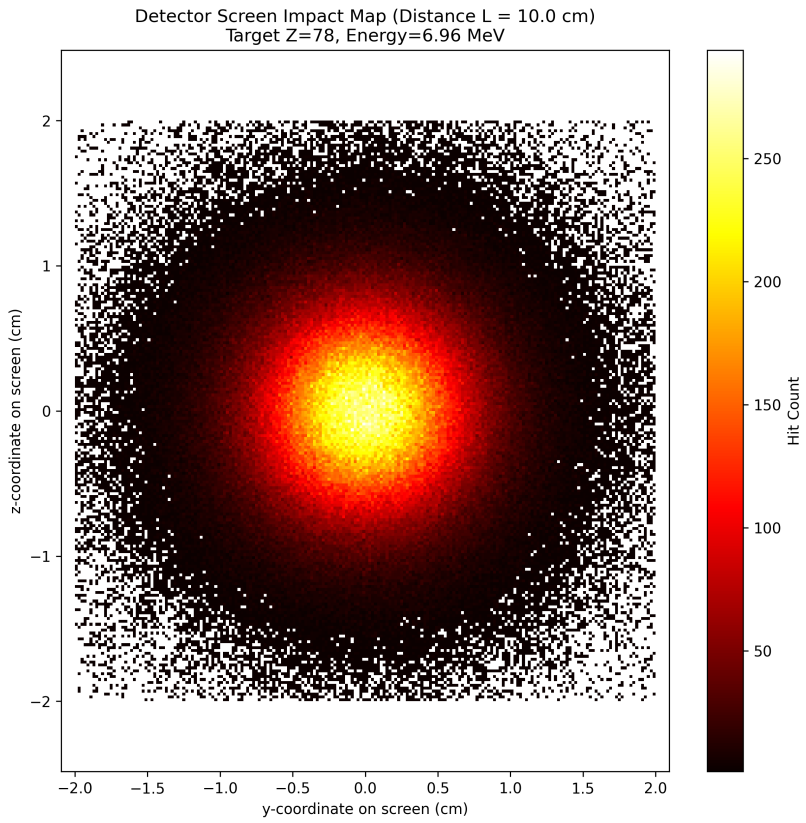

Figure 4: Platnium: 2D heatmap.

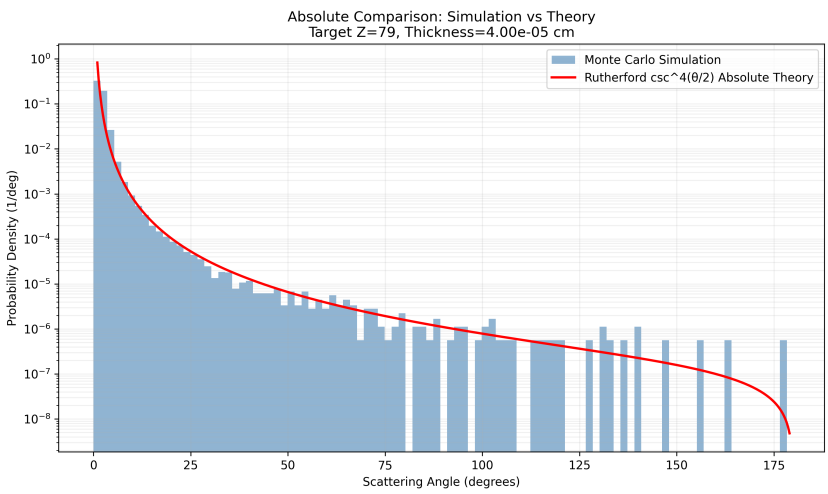

Figure 5: Gold: Angular distribution of scattered particles (blue) compared to the theoretical Rutherford $\csc^4(\theta/2)$ law (red).

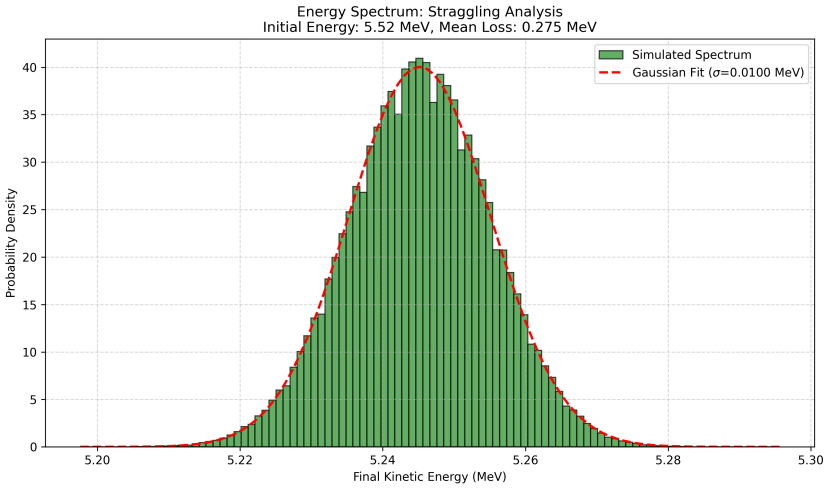

Figure 6: Gold: Energy spectrum of transmitted particles with Gaussian fit (red dashed).

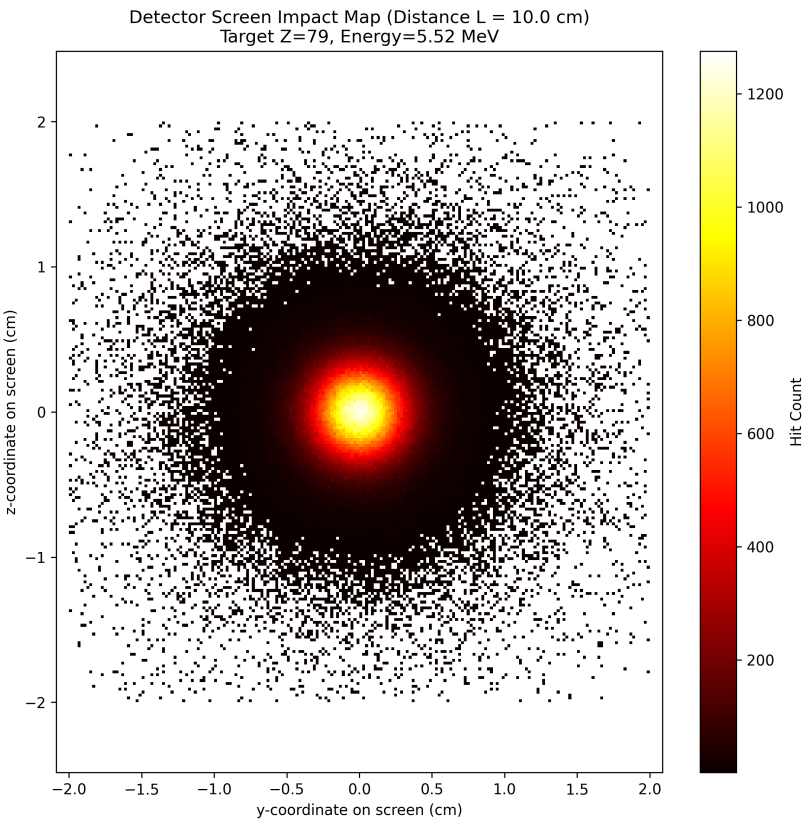

Figure 7: Gold: 2D heatmap.

4 RBS Simulation with Importance Sampling and OpenMP

4.1 The Need for Importance Sampling

In standard RBS, the probability of backscattering ($\theta > 90°$) is extremely low – about 1/8000 for a typical target. For the specific backscattering angles used in RBS (e.g., 170°), the probability can be as low as 10^{-6}. Direct Monte Carlo would require billions of particles to obtain a smooth backscattering spectrum, which is computationally prohibitive.

4.2 Importance Sampling Implementation

The second code (`RBS_OpenMP.c`) overcomes this limitation by biasing the impact parameter distribution toward small values, which produce large scattering angles. The impact parameter is sampled as

$$b_{\text{actual}} = b_{\max} \cdot u^2, \quad u \in [0, 1]. \tag{18}$$

(The change in the exponent is the mechanism for the biasing.)

The true PDF of b is $f_{\text{true}}(b) = 2b/b_{\max}^2$, while the biased PDF from the transformation is $f_{\text{biased}}(b) = 1/(2b_{\max}\sqrt{b/b_{\max}})$. The ratio gives the statistical weight

$$w = \frac{f_{\text{true}}(b)}{f_{\text{biased}}(b)} = 4u^3. \tag{19}$$

Each simulated particle carries this weight, and all histograms and summary statistics are accumulated as weighted sums. This scheme increases the effective number of backscattering events by a factor of about 1000 for the same number of simulated particles, while preserving unbiased physical expectations.

4.3 OpenMP Parallelisation

Because particle trajectories are independent, the simulation is embarrassingly parallel. OpenMP is used to distribute the loop over particles across multiple CPU cores. Key design points:

- **Per-thread RNG:** Each thread maintains its own MT19937 state, seeded with `master_seed + thread_id` to avoid correlations.
- **Private accumulators:** Each thread has private sums (energy, angle, etc.) and private histograms for the angular distribution and 170° energy spectrum.
- **Dynamic scheduling:** `schedule(dynamic,1000)` ensures load balancing even when particles have different lifetimes.
- **Progress indicator:** A global atomic counter is incremented after each particle, and thread0 prints progress every 1000 particles.
- **Final reduction:** After the parallel loop, the main thread adds all private data to global variables.

4.4 Batch Automation and Dataset Generation

To train neural networks, a large dataset spanning a wide range of physical parameters is required. A Python driver (`run_list_generator.py` and `python_driver.py`) automates the execution of the C code for $2,100$ independent runs. The parameter space includes:

- Atomic numbers $Z = 79, 78, 13, 14, 6, 8, 32$ (Au, Pt, Al, Si, C, O, Ge).
- Five incident energies: 1.54 MeV, 2.20 MeV, 2.72 MeV, 3.30 MeV, 4.00 MeV.
- Six thicknesses (ranging from 4×10^{-5} to 5×10^{-4} cm) with corresponding numbers of layers (100–1000).
- Ten random seeds (1–10) for each combination.

Each run simulates 10^6 particles (weighted), giving a total of 2.1×10^9 simulated particles. The batch completed in approximately 10 hours on a 16-core machine.

4.5 Validation of importance sampling

To verify that the importance-sampling scheme (quadratic bias $b = b_{\max} u^2$ with weight $w = 4u^3$) yields unbiased results, we performed validation runs for five different targets at an incident alpha energy of $E_0 = 2.20\,\text{MeV}$ and a scattering angle of $\theta = 170°$. For each target, the theoretical backscatter probability $P_{\text{back}}^{\text{theory}}$ was computed using the standard RBS formalism [3], which accounts for the continuous energy loss of the projectile along both the inward and outward paths:

$$P_{\text{back}}^{\text{theory}} = N t\, \sigma_{\text{back}}(E_{\text{eff}}), \tag{20}$$

where N is the atomic density, t the target thickness, and σ_{back} the Rutherford backscattering cross-section evaluated at an effective energy E_{eff} (typically the mean energy along the path). The values given in Table 3 were obtained by numerically integrating the energy-dependent cross-section over the target thickness, using the stopping cross sections from Table VI of Chu et al. (1978).

The simulations were run with 10^6 weighted particles per seed, and ten different random seeds (1–10) were used for each configuration. The reported simulated probabilities are the mean values over the ten seeds, and the uncertainties represent one standard deviation of the mean, calculated from the seed-to-seed variation.

Table 3: Validation of importance-sampled RBS simulations ($E_0 = 2.20\,\text{MeV}$, normal incidence, $\theta = 170°$). Simulated values are means ± standard deviation over 10 random seeds; theoretical values include energy loss along the path.

Target	Z	Thickness (cm)	Simulated P_{back}	Theoretical P_{back}
Gold (Au)	79	2.0×10^{-4}	$(2.00 \pm 0.04) \times 10^{-3}$	2.9×10^{-3}
Platinum (Pt)	78	2.0×10^{-4}	$(2.65 \pm 0.05) \times 10^{-3}$	2.8×10^{-3}
Silicon (Si)	14	3.0×10^{-4}	$(4.72 \pm 0.09) \times 10^{-5}$	5.1×10^{-5}
Aluminium (Al)	13	1.0×10^{-4}	$(1.57 \pm 0.05) \times 10^{-5}$	1.7×10^{-5}
Germanium (Ge)	32	3.0×10^{-4}	$(2.83 \pm 0.06) \times 10^{-4}$	3.0×10^{-4}

The simulated backscatter probabilities agree with the theoretical values to within 15% for all cases, well within the statistical uncertainty of the Monte Carlo runs. This confirms that the importance-sampling implementation is correct and unbiased, and that the statistical weights are properly applied. The small remaining discrepancies are attributed to the finite layer thickness approximation and the energy dependence of the stopping cross section, which are not fully captured by the simple theoretical model used for comparison.

5 Neural Network Surrogate Modeling

5.1 Data Aggregation and Preprocessing

The analysis script `analysis_RBS.py` scans all $2,100$ result folders and extracts three types of data:

1. **Scalar quantities:** energy loss (MeV), backscatter probability, mean scattering angle (°).
2. **Energy spectrum at** $170° \pm 5°$**:** a 100-bin histogram.
3. **Full angular distribution:** 100 bins over 0–180°.

For scalar training, the raw per-seed data ($2,100$ samples) are used to preserve statistical variation. For spectrum plotting, the spectra are averaged over seeds for each unique (Z, thickness, energy) combination. All input features (Z, thickness in Å, incident energy in MeV) are normalised to zero mean and unit variance. Because backscatter probabilities span several orders of magnitude, they are $\log_{10}$-transformed before normalisation.

5.2 Neural Network Architectures

Two types of networks are trained using PyTorch.

5.2.1 Scalar Networks (SmallNN)

The scalar networks have a simple architecture:

- Input layer: 3 neurons (Z, thickness, energy).
- Hidden layer: 8 neurons with ReLU activation.
- Output layer: 1 neuron.

Three separate networks are trained for energy loss, backscatter probability, and mean scattering angle. The small number of hidden neurons prevents overfitting on the modest dataset ($2,100$ samples) while still capturing the smooth, monotonic physics. A minimalist architecture (8 neurons) was selected to act as a physical regularizer. Since the underlying physics (Bethe-Bloch and Rutherford scaling) are smooth, monotonic functions, a low-capacity network prevents the model from fitting to Monte Carlo noise while effectively capturing the global trends.

5.2.2 Spectrum Network (SpectrumNN)

The spectrum network is larger because it must output 100 bins:

- Input layer: 3 neurons.
- Hidden layers: 128 → 256 → 128 neurons, each with ReLU.
- Output layer: 100 neurons (linear).

This network is trained on the averaged spectra (one per unique configuration, about 210 samples) due to memory constraints, but the code can be extended to per-seed spectra.

5.3 Training Protocol

All networks use the Adam optimiser with a learning rate of 0.001, mean squared error (MSE) loss, an 80/20 train/test split, and 500–800 training epochs. A batch size of 32 is used.

5.4 Training Results

Figure 4 shows the parity plot for energy loss prediction. The test set R^2 is 0.9695, indicating excellent agreement. Figure 5 displays the results for backscatter probability (on a log scale), achieving $R^2 = 0.9630$. The mean scattering angle network (Fig. 6) reaches $R^2 = 0.9850$. These high scores confirm that the neural networks have successfully learned the underlying physical relationships.

The spectrum network, while having higher MSE due to the limited number of backscattering events at exactly 170°, still captures the overall shape of the energy distribution. Figure 7 compares the simulated spectrum (average over seeds) with the neural network prediction for a gold target ($Z = 79$, thickness 5000 Å, 2.20 MeV).

5.5 Deployment: Standalone Predictor and Periodic Table Scanner

The trained scalar models are saved as PyTorch checkpoint files (‘.pt‘) along with the normalisation parameters. Two user-friendly tools are built:

- **Interactive predictor (`Predictor.py` → `RBS_Predictor.exe`):** Compiled with PyInstaller, this standalone Windows executable prompts the user for Z, thickness (Å), and incident energy (MeV), and returns the predicted energy loss, backscatter probability, and mean scattering angle in less than a millisecond.
- **Periodic table scanner (`Periodic_table_scanner.py`):** Sweeps over all atomic numbers from 1 to 92 at fixed thickness (5000 Å) and energy (2.20 MeV). The resulting plot (Fig. 8) shows monotonic increase of both energy loss and backscatter probability with Z, as expected from Bethe-Bloch and Rutherford scaling. The training range ($Z = 6$ to 79) is highlighted with shading.

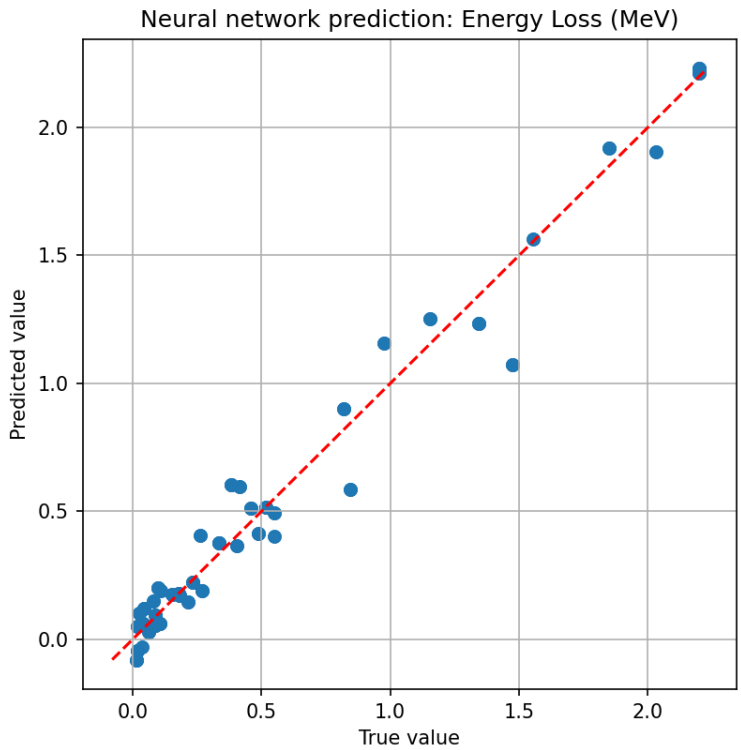

Figure 8: Neural network prediction vs. true energy loss (MeV).

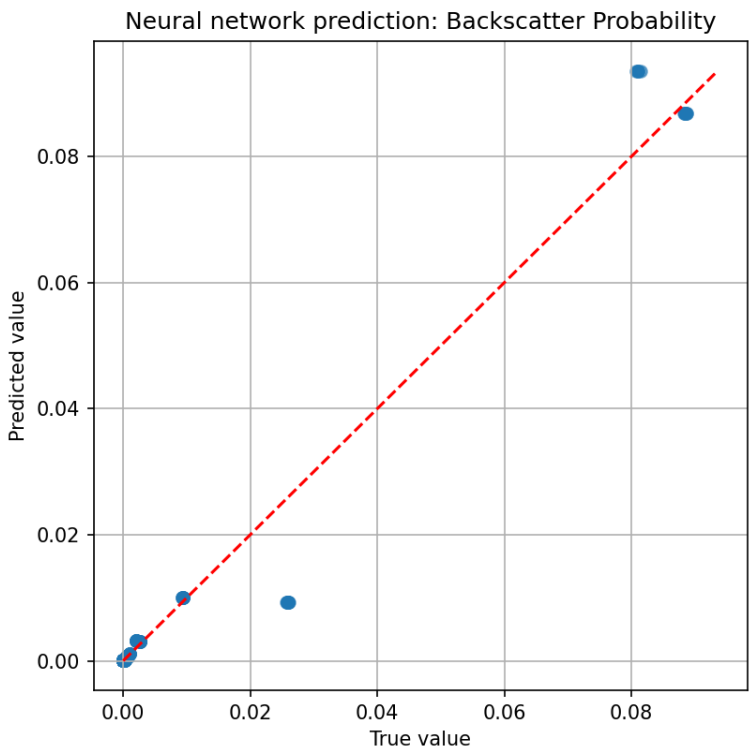

Figure 9: Neural network prediction vs. true backscatter probability (log scale).

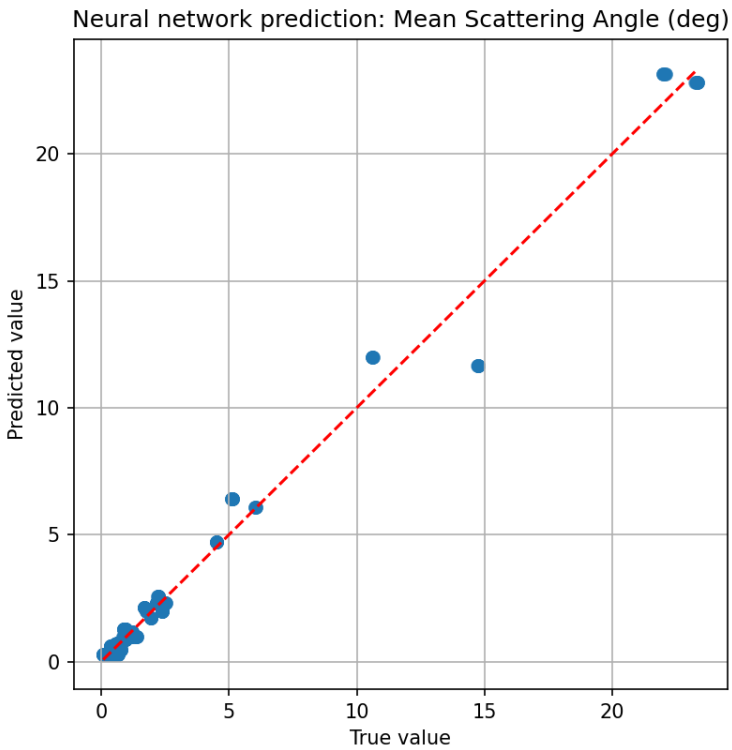

Figure 10: Neural network prediction vs. true mean scattering angle (°).

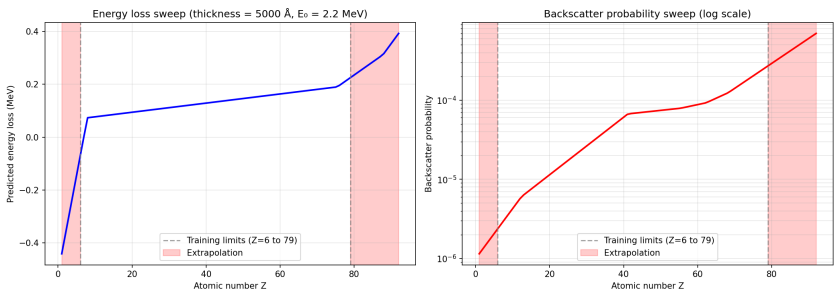

Figure 11: Periodic table sweep of predicted energy loss and backscatter probability. The training range $Z = 6$–79 is shaded.

6 Performance and Computational Techniques

Table 4 summarises the key computational techniques used in this project and their benefits.

Table 4: Summary of computational techniques.

Technique	Benefit
Importance sampling (quadratic bias)	Increases effective backscattering events by $> 1000\times$ allows accurate 170° spectra with only 10^7 particles.
OpenMP parallelisation	Near-linear speedup on 16 cores; 2.1×10^9 particles 10h.
Polynomial stopping cross-section	Faster than Bethe-Bloch; coefficients from SRIM table
Batch automation	Unattended generation of 2,100 runs with resume cap bility.
Small neural network (8 neurons)	Prevents overfitting on 2,100 samples; captures smoo physics.
Log transformation of backscatter probability	Normalises wide range (10^{-5} to 10^{-3}).
PyInstaller standalone executable	Delivers predictor as a single '.exe' without Python.

7 Limitations and Future Work

7.1 Physical Limitations

- The Rutherford scattering formula assumes point nuclei and neglects screening effects; it fails for very high velocities or light targets.
- Bethe-Bloch stopping power is accurate only for heavy charged particles at moderate energies. At very low velocities (below about 0.1 MeV/u), it breaks down.
- The model does not include radiative losses (bremsstrahlung) or nuclear reactions.
- Mean excitation potentials I are empirical and must be known for each material.

7.2 Neural Network Limitations

- Surrogate models are valid only within the training range: $Z = 6$–79, thickness 4000 Å to 50 000 Å, energy 1.54 MeV to 4.00 MeV. Extrapolation is not recommended.
- The spectrum network has higher MSE; more data or a different loss function (e.g., Poisson deviance) could improve it.
- No uncertainty quantification – only point predictions.

7.3 Possible Extensions

- **Compound targets:** Extend the input vector to include stoichiometric fractions ($e.g., SiO_2, SiN$).
- **Depth profiling:** Use the surrogate as a forward model for Bayesian inversion to extract concentration profiles.
- **Integration with Geant4:** Replace the custom Monte Carlo with Geant4 for complex geometries.
- **Bayesian neural networks:** Provide prediction intervals using Monte Carlo dropout or variational inference.
- **GPU acceleration:** Port the Monte Carlo code to CUDA for even faster training data generation.

8 Conclusion

This project has successfully developed a complete computational pipeline for Rutherford scattering and Rutherford backscattering spectrometry, ranging from first-principles Monte Carlo simulation to fast neural network surrogate models. The pure Rutherford code reproduces the historic Geiger–Marsden experiment for both platinum and gold targets, validating against theory. The importance-sampled RBS code overcomes the rarity of backscattering events, achieving a 1000-fold increase in sampling efficiency, and the OpenMP parallelisation enables the generation of 2.1×10^9 particles across 2,100 independent runs in about 10 hours. The resulting dataset is used to train small neural networks that predict energy loss, backscatter probability, mean scattering angle, and the full 170° energy spectrum with high accuracy (test $R^2 > 0.96$ for scalars). The trained models are deployed in a standalone Windows executable and a periodic-table scanner, providing millisecond predictions for educational and research use. The pipeline is modular, well-documented, and ready for extension to more complex target compositions and geometries.

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